

The Impact of Different Obstacle Outline Enhancement in Augmented Reality on Walking Experience

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ABSTRACT

Improving obstacles visibility can improve mobility, a significant challenge for visually challenging scenarios or with visual impairments. As a first step in developing such a system, we examined the experience of walking an obstacle course in typically sighted participants across four conditions – passthrough, partial augmentation (physical obstacle outline was augmented), full augmentation (virtual obstacles were fully overlaid on the physical obstacle), and virtual-only objects. Walking time was significantly slower in the partial condition. Interestingly, although the visibility of obstacles differed across augmentations, and affected walking speed, participants did not notice this difference and rated the various augmentations similarly.

Index Terms: Augmented Reality, Navigation, Accessibility

1 INTRODUCTION

Mobility and safety during walking is a critical part of independent living and quality of life. Our long-term goal is to enhance the visibility of obstacles by highlighting the outline of upcoming obstacles with Augmented Reality (AR) thereby increasing the ability to detect them in real time while walking. Enhancing the visibility of obstacles can help older adults who face age related declines in visual perception [1], people with low vision [2, 3], as well as typically sighted people encountering situational impairments such as being distracted or using their phone while walking [4]. Improving the detection of obstacles can improve safety, comfort, and prevent collisions. The potential of AR technologies to directly enhance vision for improving daily life tasks has been demonstrated in prior research for example by assisting the obstacle detection for people with low vision, improving their spatial awareness, and increasing users' confidence in mobility [5].

Here, we developed an application that renders various types of designs of AR enhancements of obstacles. The AR enhancements are based on outlining the shape of obstacles with virtual highlights, keeping the appearance of obstacles as similar as possible to their real-world appearance, thus allowing a smooth experience. The enhancements differed in their combinations of real versus virtual elements. As a first step, in the current study we examined the feasibility of our application, enhancement designs, and the walking experience in typical participants focusing on walking time, collisions, and subjective experiences (Fig. 1).

2 METHODS

Hardware: We used the Meta Quest Pro headset.

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Figure 1: Experimental setup and conditions. Left to right - Passthrough; VR only, obstacles were entirely virtual; AR-Full, real obstacles were fully overlaid with virtual representations; and AR-Partial, outline of real obstacles were highlighted in AR.

Software: Daily obstacles that were present in the hallway were scanned using an iPad with a depth sensor. Then, the outline of each obstacle was drawn by a graphic designer in Blender. The Unity application included positions/orientations of objects and recorded user's position, eye, head movements, and first-person video.

Ethics statement: The experiment was approved by the ethics committee of the University of Haifa (approval 085/22).

Participants: Eleven participants with no prior experience walking with VR headsets, aged 24-35 (6 females), with typical vision.

Experiment Procedure: Each participant signed a consent form and completed a demographic questionnaire. Before the experiment started, participants adjusted the headset for their comfort. Each participant practiced walking with the glasses in an empty hallway without any obstacles and a hallway with virtual animals to get used to wearing and walking with the headset. Eye movement calibration was performed using the Meta Pro headsets. The obstacle course included fifteen daily-life obstacles, such as chairs and tables along a 15-meter office hallway. The study consisted of four conditions in passthrough mode (viewing themselves and the real hallway) that only varied in obstacle AR manipulation: (1) Virtual obstacles (VR only) - Virtual obstacles displayed in passthrough mode in the real corridor (with no physical obstacles); (2) Passthrough: Real obstacles with no virtual AR overlays; (3) Opaque obstacle augmentation of real objects (AR-full) - Opaque virtual overlays of obstacles with pink outlines were accurately superimposed onto real obstacles; (4) Transparent augmentation of real obstacles (AR-Partial) - Partially transparent virtual overlays of obstacles with pink outlines were accurately superimposed onto real obstacles. The obstacles were positioned differently for each condition in a counterbalanced design to minimize order effects. Participants wore the headsets in each condition and were blinded to the experimental condition. Obstacle locations on the path were pre-set for each condition. During the experiment, the researcher positioned each real obstacle in place according to these pre-set locations and the AR overlays were examined by the researcher to ensure accurate overlap between physical and augmented features. In each condition, participants walked the course back and forth. Participants had no prior knowledge of the obstacle layout; they had to pass through a virtual wall that blocked their view of the course, revealing the obstacles only after crossing.

Dependent Measures: We measured the time it took each participant to walk the course back and forth using a timer. In one

condition for one participant, the timer did not work correctly, and the time was calculated from the application's log file. Participants completed a questionnaire, rating statements between 1-7 (strongly agree). Finally, we conducted semi-structured interviews to gain insights about their experiences and the visibility in each condition.

3 RESULTS

3.1 Walking Time

We performed a repeated measures ANOVA to assess the effects of different conditions on walking times. The results showed a significant effect of condition on walking time ($F(3, 10) = 4.863$; $p < 0.01$; $\eta^2 = 0.32$), indicating notable differences in average walking times across conditions. Post hoc comparisons using the Holm correction revealed that walking time in the AR-Partial condition was significantly slower than both the AR-Full condition ($t(10) = 3.37$; $p < 0.05$; $Cohen's d = 0.89$) and the Passthrough condition ($t(10) = 3.22$; $p < 0.05$; $Cohen's d = 0.93$). No other comparisons were significant, showing that AR-Full and Passthrough conditions did not significantly differ.

3.2 Walking Experience

Questionnaire: Participants were asked to rate their experiences on various aspects after completing each condition. The ratings included confidence in navigating, awareness of obstacles, clarity of the visual display, and the perceived effectiveness of the system. Overall, there was no clear preference for any specific condition, as participants rated their experiences similarly across all conditions ($p > 0.1$, $\eta^2 < 0.002$). This suggests that participants did not experience the course in a different way despite the different augmentations and the difference in walking speed.

Semi-structured interviews: Overall, participants did not describe a different experience between the three types of enhancement conditions, "In terms of walking and spatial awareness, honestly, I didn't feel any difference" (P2). Most participants did not express a clear preference for a specific enhancement, describing similar feelings of comfort and safety across all conditions. For example, "I didn't have a real preference; they all felt about the same to me" (P11). Some participants preferred the passthrough condition, which involved an obstacle course without any AR additions. They indicated that seeing the course as it was, without additional markers, contributed to a greater sense of control during navigation, e.g., "When I saw the course as it is, it felt more natural and easier to navigate" (P10). Participants noted that the AR obstacle highlights did not always assist them, and were sometimes confusing. Some participants reported that it negatively affected spatial awareness: "The marker was so prominent that it just confused me – it would have been better without it." (P2). We asked participants if they noticed walking in a condition without real physical obstacles, participants did not notice that no real obstacles were present on the course. One participant said: "I didn't realize I wasn't walking in a real obstacle course. It looked very real to me." Finally, referring to potential use cases, participants noted that highlights could be used to attract attention to objects, e.g. "Working with children on attention issues, using such highlights could help them focus on relevant versus irrelevant details, and it could be really useful." (P9).

4 DISCUSSION AND FUTURE WORK

Our results showed that participants walked slower in the AR-Partial condition compared to AR-Full and passthrough conditions, indicating that the partial overlays may have introduced challenges in navigating the obstacles. These results suggest that the AR-full condition is better suited for future systems to assist obstacle enhancement and walking. At the same time, the subjective ratings indicated no preference between AR conditions, suggesting a similar overall experience. Interestingly, participants did not notice that one condition contained no real obstacles (VR only), suggesting that

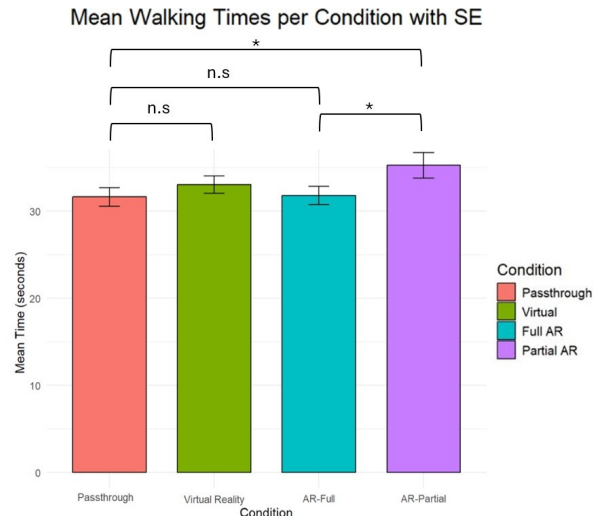


Figure 2: Results for time across conditions. Errors are standard error of the mean.

under mixed reality situations, virtual obstacles which are realistic enough would be treated the same as physical ones. This aligns with findings that VR training can enhance collision-avoidance behavior in older adults [6]. This has practical implications for utilizing virtual objects in rehabilitation protocols, and for the potential criticisms of studies utilizing virtual objects only as not reflecting naturalistic user behavior. A limitation of our study is that we tested our system on participants with typical vision and in scenarios without visual challenge – future work will need to include users with visual challenges (low vision or older adults) or during visually challenging scenarios (e.g., walking and texting) who may benefit more significantly from visual augmentations. To characterize user behavior additional measures such as eye and head movements as well as walking properties could be analyzed in the future. Participants also suggested that AR enhancements could be beneficial for attention training for children, particularly for focusing on relevant versus irrelevant details.

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